Turtle Games — Sales Improvement Project (Report)

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1) Business objectives

Main Business Objective:

Improve Turtle Games' **sales performance** by utilizing **customer analytics** to inform marketing & targeting strategies.

Supporting Objectives:

- **Loyalty Point Engagement:** Identify main drivers for loyalty_points & how well we can model & predict them
- **Customer segmentation**: Explore ways to segment customers to implement effective marketing (acquisition and retention) strategies
- **Customer Sentiment:** Utilize review sentiments to understand satisfaction & potential churn risk and thus inform marketing decisions

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2) Data and preparation

Dataset:

Turtle Games customer dataset contains –

- *loyalty_points* - **predicted variable,**
- *spending_score, remuneration, age* - **main numerical predictors,**
- *education* & *gender* - additional categorical variables,
- *review* & *summary* - **text**

Data ingestion, manipulation & wrangling were performed using Python (numpy, pandas, scikit-learn etc.) and R (dplyr/ggplot2), followed by a data preparation workflow below to convert **raw Turtle Games data** into **analysis-ready versions** for:

- Predictive Modelling/Clustering → Step 1 + Step 2
  &
- Sentiment Analysis→ Step 1 + Step 2 + Step 3

Step	Action
1	Separated structured vs text workflows: numeric/categorical fields for modelling and clustering; review/summary fields for NLP.
2	Missing values and duplicates validation (e.g., sum(is.na), duplicate checks) and enforced correct data types (e.g., factors for categories).
3	Normalization (case and punctuation handling), lemmatization, alphanumeric and duplicate text removal to avoid unwanted noise in sentiment outputs.

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3) Exploratory analysis and its suggestions

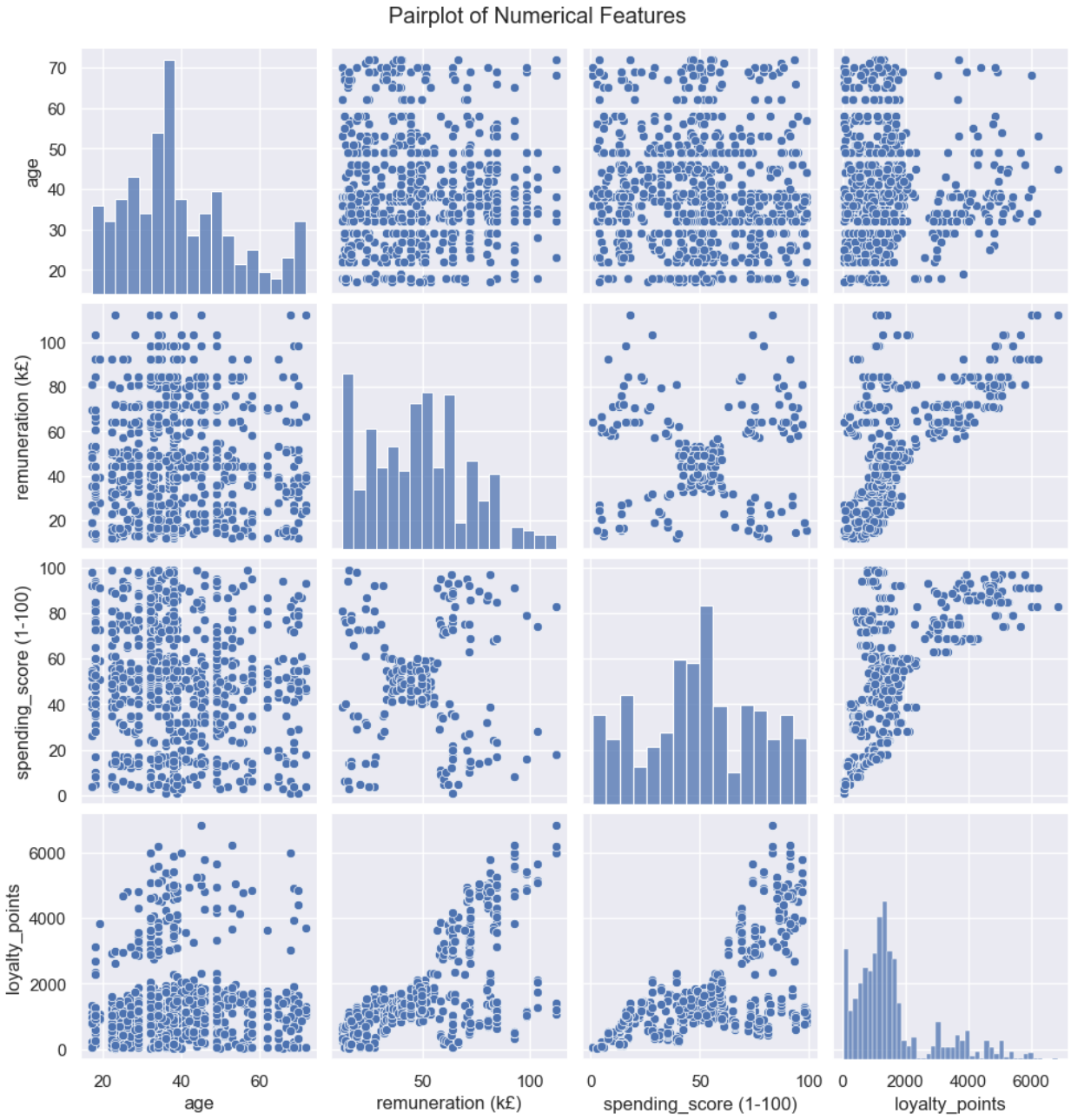
Summary statistics, distribution/correlation plots, graphical aids & cluster analysis were utilized to reveal patterns, groups, and predictability.

3.1 Structured EDA

- ***Loyalty_points*** have a **right-skewed** distribution and showed a **weak secondary mode**

  → Evidence of **multiple subgroups of customers** (some customers accumulate very high loyalty).

- ***Spending_score*****,** ***remuneration***, show a **higher association** with ***loyalty_points****,* than *age.*



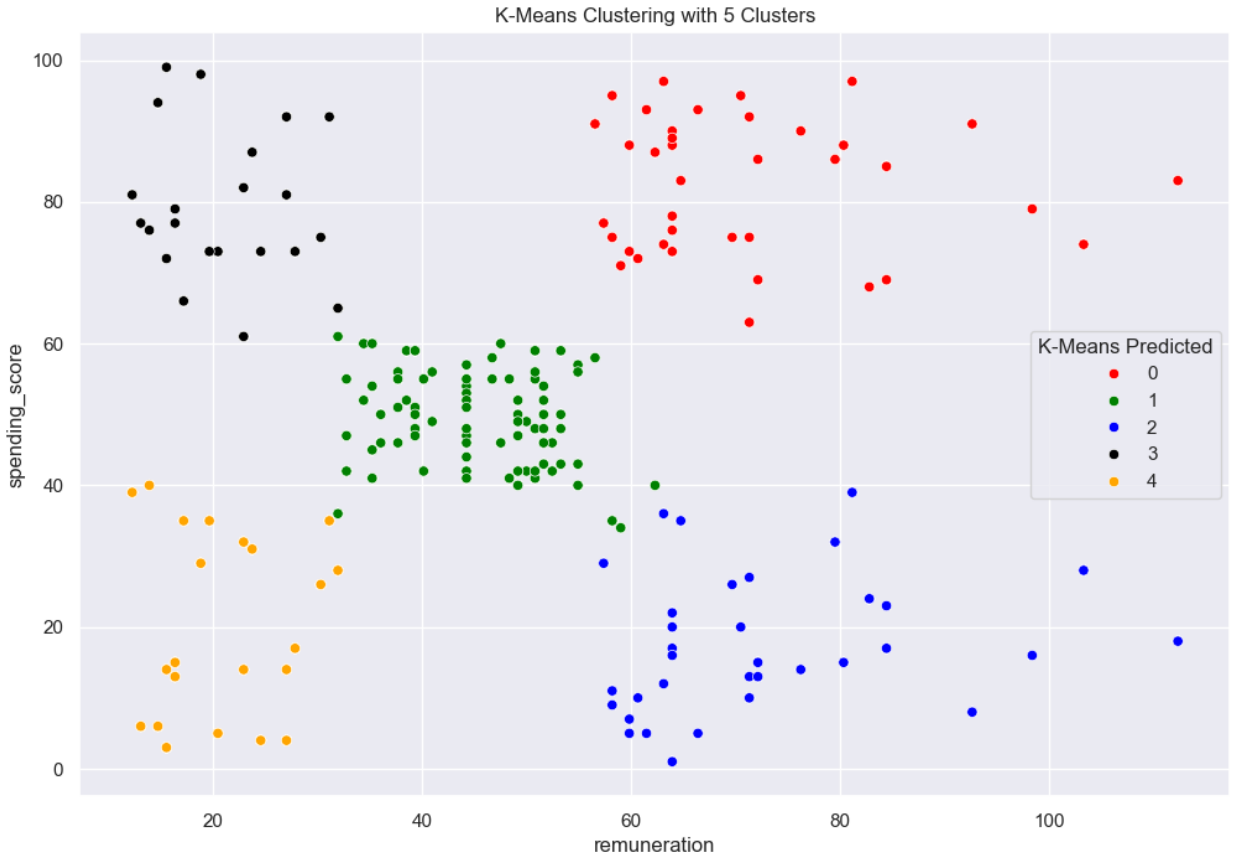
- Overall, ***Loyalty_points*** increases with spending_score and remuneration, but the relationship breaks down at higher spending/remuneration ranges, where **numeric predictors do not** exhibit a **perfectly linear relationship.**

  → **Non-linear models** should be considered.

3.2 Clustering Exploration:

Applied K-means on spending_score and remuneration selected **k=5** (elbow + silhouette) & observed:

- Low remuneration splits into **low-spend** and **high-spend** subgroups.
- High remuneration also splits into **low-spend** and **high-spend** subgroups.
- A mid remuneration / mid spending segment exists between them.

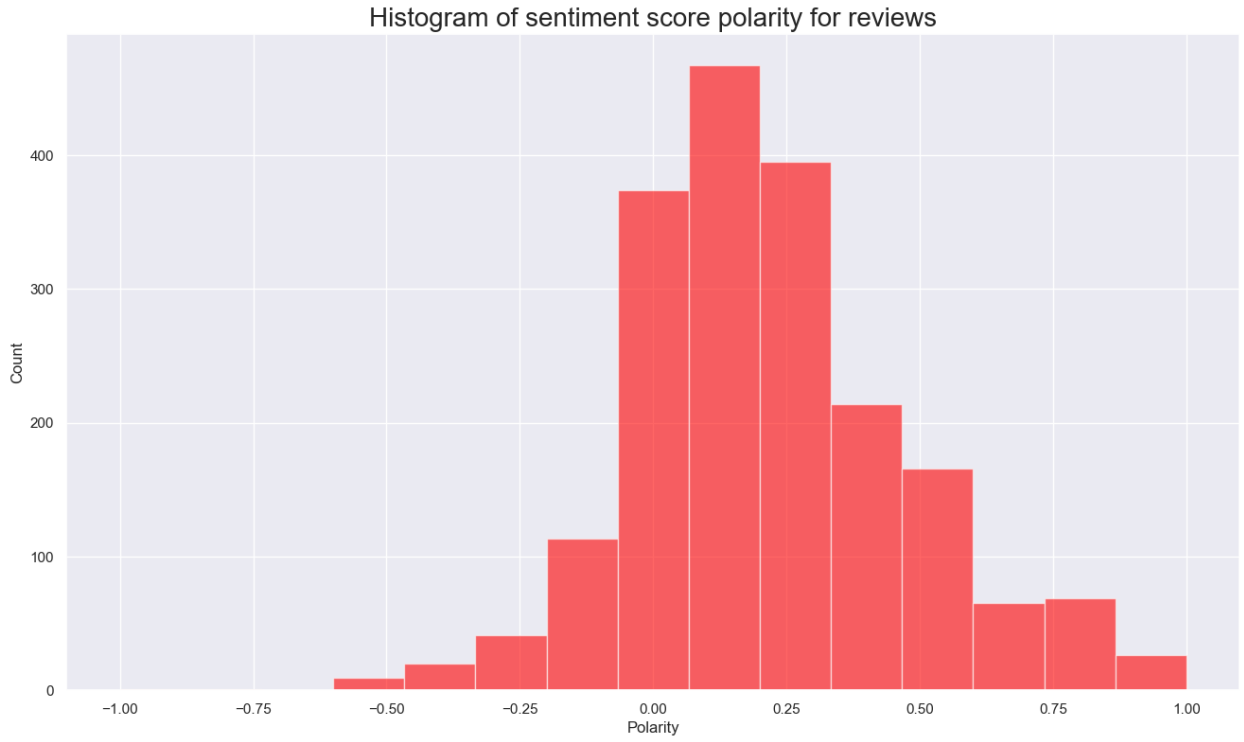


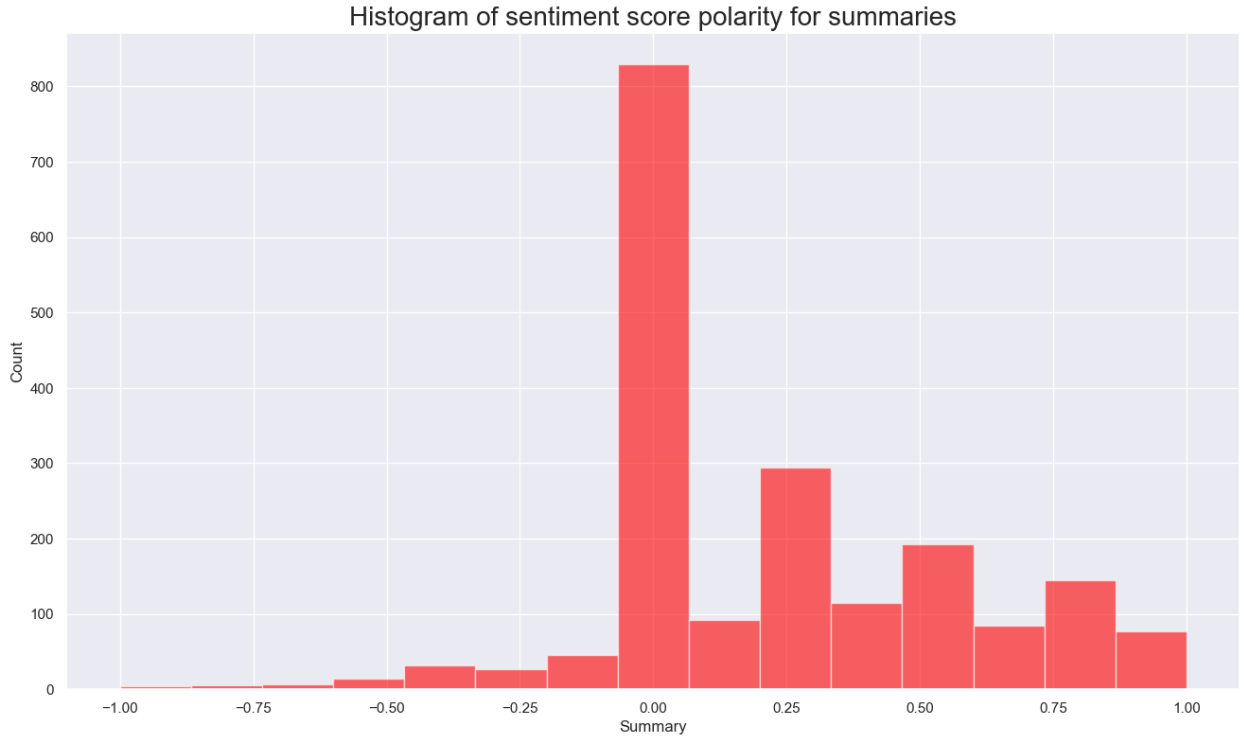
3.3 Text Exploration (sentiment):

TextBlob scored sentiment for reviews & summaries.

- An overall positive sentiment score was observed across reviews & summaries, with a slightly larger neutral sentiment in summaries. The summaries column has most entries with ellipses & TextBlob captures most of them as neutral. **Review texts are more reliable for sentiments.**
- Wordcloud for negative sentiments is shown below. It was observed that TextBlob is a weak sentiment predictor model, often **missing the classification of rhetorical patterns correctly**.

  → Textblob-driven sentiment analysis is a good reference point; however, not entirely accurate.


Histogram of sentiment score polarity for reviews
Count
0
100
200
300
400
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Polarity


Histogram of sentiment score polarity for summaries
Count
0
100
200
300
400
500
600
700
800
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Summary



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4) Model Selection & Findings

Modelling Procedure & Assessment:

All candidate models were built to predict ***loyalty_points*** using the available predictors (Section 2), with a train/test split to estimate out-of-sample error, and compared across:

- **Predictive capability:** RMSE, MAE, R^2
- **Model diagnostics:** Residual structure, Shapiro (normality), Breusch–Pagan (heteroskedasticity), and VIF (multicollinearity)
- **Non-linear explainability/stability:** Feature importance, permutation importance (ΔMAE), and bootstrap variability (ranking stability)
- **Overfitting controls:** predictors were added **one-by-one** in linear models (retain only if test error improved / diagnostics stabilized); tree models were **pruned** via max_depth / min_samples_leaf to balance accuracy and generalization

Model Comparison:

1. Linear Model Iterations

Model	Type	Predictors	Test RMSE	Test MAE	R^2 (test summary)
MLR-1	Linear	spending_score + remuneration + age	548.2729	427.6864	0.8381
MLR-2	Linear - Age Squared	spending_score + remuneration + age + age^2	518.5747	374.4682	0.8512
Log1p target MLR	Linear-log(target)	spending_score + remuneration + age + age^2	1033.196	553.3235	0.8231

2. Non-Linear Model Iterations

Model	Model family	Predictors	Test RMSE	Test MAE	R^2 (test summary)
Decision tree regressor (tuned)	Tree-based	spending_score,remuneration, age	228.479	137.373	0.969

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5) Results & Diagnostics

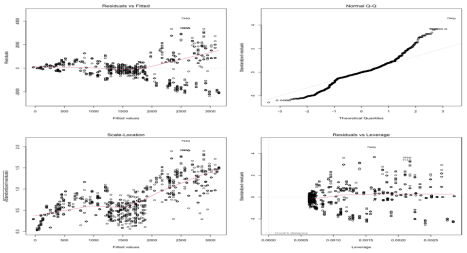
Linear Modelling

1. Residuals indicated **misspecification**: curvature, **non-normality** (low Shapiro p), and **heteroskedasticity** (low Breusch-Pagan p); predictions could also turn **negative** (unconstrained regression).

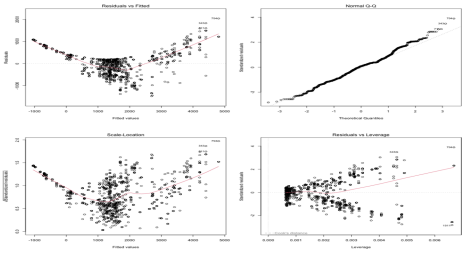
    - Adding **age²** improved fit/accuracy.
    - **Log(*loyalty_points*)** removed negative outputs, but reduced accuracy.

→ Despite these adjustments, **nonlinear residual structure persisted**, suggesting linear form limits for this dataset.

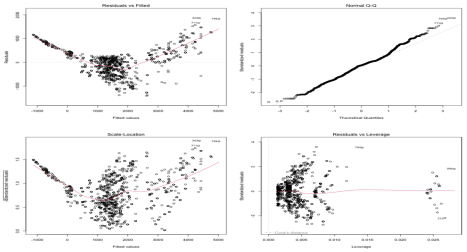
spending_score



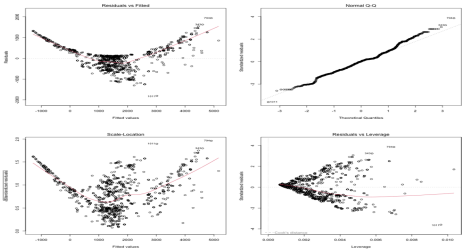
spending + remuneration



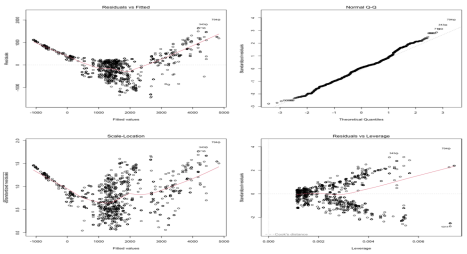
spending_score + remuneration + education



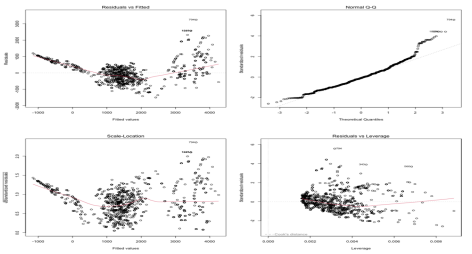
spending_score + remuneration + age



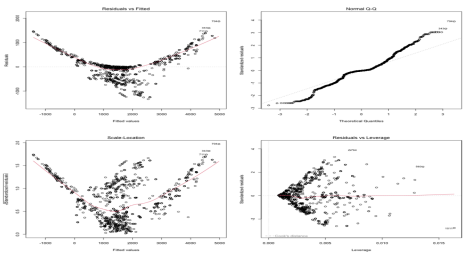
spending_score + remuneration + gender



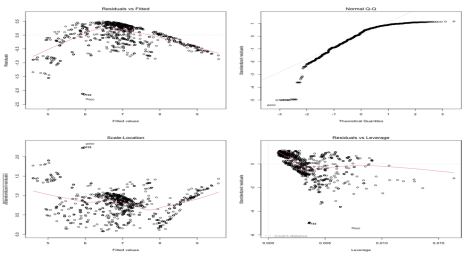
spending_score + remuneration_band + age

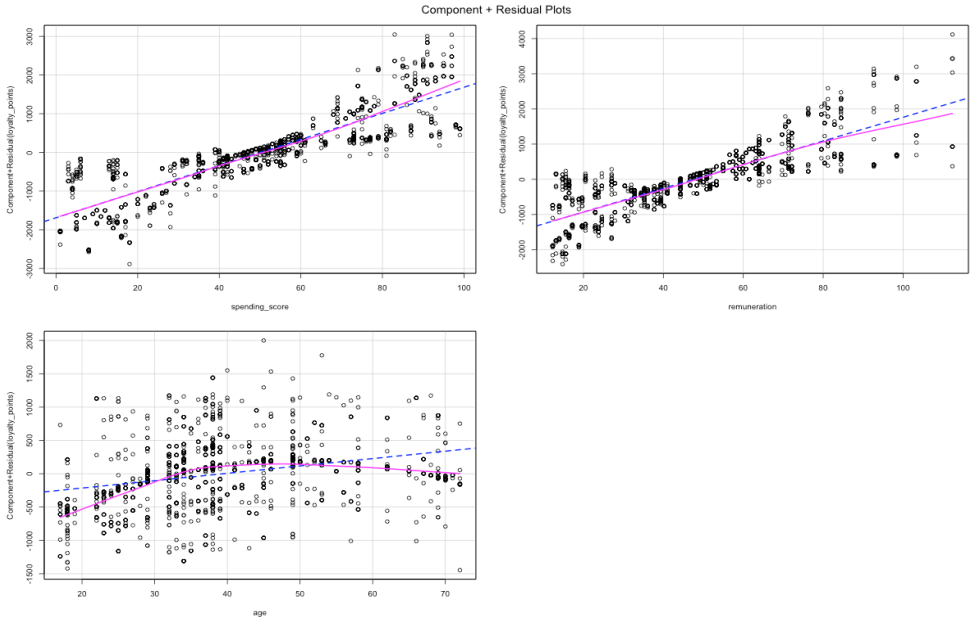


spending_score + remuneration + age + age^2



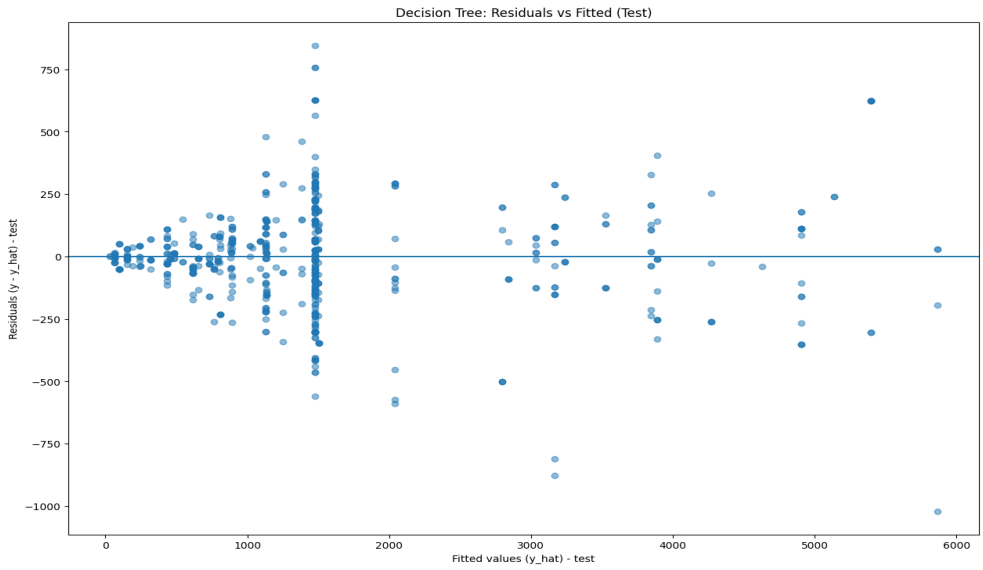
log version





Non-Linear Modelling

1. **Decision Tree Regressors'** residuals showed stepwise vertical stripes (common in tree models). Error variance is **uneven across fitted ranges**, indicating **Heteroskedasticity**.



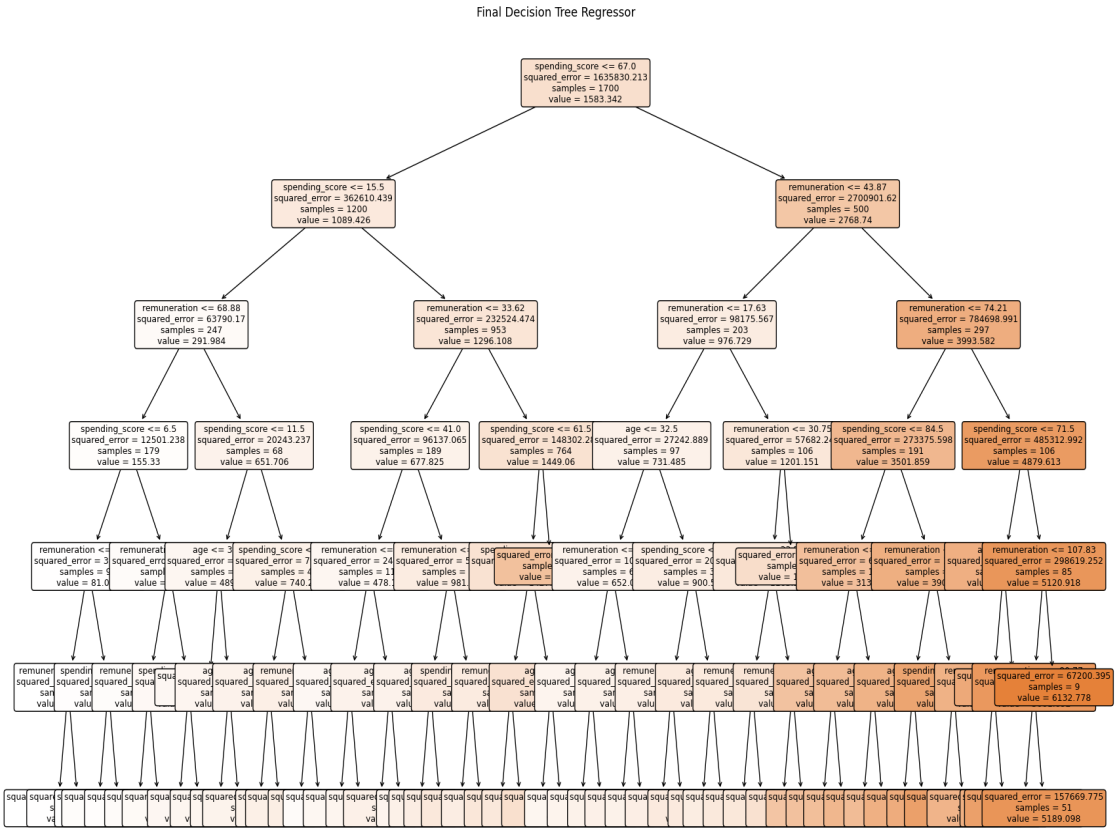
2. **Bootstrapped-feature-importance**, showed overall stable importance ranking & **overlapping feature** importance between **spending_score** (0.50945 ± 0.053) & **remuneration** (0.48015 ± 0.054), implying either can drive nodesplit at any depth, without dominating each other.

3. **Cross-validated permutation importance** (ΔMAE) also shows that **spending_score** and **remuneration** are the **dominant drivers** of out-of-sample accuracy (largest |ΔMAE| when features are included/excluded), while age has a marginal effect.

→Overall, the tree captures **non-linear structure** better than linear models, but error variance differs by segment.

Results

- **Best linear model (interpretable baseline):**
    - **MLR with age²:** spending_score + remuneration + age + age²
- **Best overall model (final choice):**
    - **Decision Tree Regressor (tuned)** using spending_score, remuneration, age



6) Business Insights

- **Loyalty Point Engagement:** *loyalty_points* is skewed and best explained using spending_score, remuneration, and age (lower incremental explainability), although the exact nature of the relationship isn't linear.
- **Segmentation**: Five spending-remuneration-based customer clusters identified, indicating **acquisition opportunities** and **clear retention priorities** for high-value segments.
- **Sentiment**: Sentiments are mostly positive; however, negative reviews suggest possible churn risk. Interpret summary sentiment cautiously since many have short/elliptical text, and Textblob falls behind in rhetoric/sarcasm.

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7) Recommendations

- Use **cluster-grouping** combined with **sentiment analysis** to **initiate triage-based activation (**untapped segments**)** and **re-engagement/retention** (at-risk) **marketing campaigns.** Validate **the performance with** conversion/retention metrics **& calibrate** periodically(monthly/quarterly):

Remuneration	Spending	Sentiment	Action Needed
High	Low	Negative	Marketing surveys to understand customer pain points Product analysis to understand product features that loses customers Initiate revised product lines and market campaigns
High	Low	Neutral/Positive	Extend marketing campaigns to help products reach customers
Low	High	Negative	Likely to lose customer base - Need immediate marketing intervention & offers
Low	High	Neutral/Positive	Target for retention-oriented marketing

			campaigns & offers

    - Secondary attention to the below remuneration/spending combo:
        1. Low-low: smaller-scale re-engagement program
        2. Mid-mid: aim to grow steadily, watching for customer preference.
        3. High-high: retention-premium rewards
- Use the **Tree model** to:
    - **Predict or validate *loyalty_points*** distribution within the **remuneration-spending-sentiment-clusters above** or otherwise obtained from **leaf-level threshold values** to further direct suitable **marketing actions**.
    - Since the data is heteroskedastic, aim to retrain the data where necessary.
- Limitations and next steps:
    - Validate with alternative clustering, as k-means assumes spherical clusters.
    - Textblob struggles with sarcasm/negation; consider a labelled validation sample and a higher-order model (VADER or transformer model),
    - Tree models can be high-variance; random forests or gradient boosting can improve stability if interpretability requirements permit.